

Exercise Sheet 4

Model Testing and Validation

pass and fail vs test metrics. Path coverage vs OOD k fold

1. all implemented functions are known in traditional testing
2. every implemented function can be tested separately in traditional testing, models can only be tested holistically
3. test results might vary drastically with small changes in the input for machine learning models

4. traditional testing tests logic and might find subsets that fail
ML tests can only test given data inputs and estimate a blackbox logic

Testing Concepts

Exercise 4.1: Traditional Testing vs. ML Model Testing

Name at least four differences between testing traditional software and testing a machine learning model.

Exercise 4.2: Test Oracles

optional

In traditional software testing, a *test oracle* determines whether a test has passed or failed.
ground truth labels!

1. What plays the role of the test oracle in ML model testing? ML Engineer and test metrics
2. Why is defining a good test oracle particularly difficult for perception tasks?
label clarity: classes can't always be well separated, ambiguous results even with domain knowledge possible
fuzzy domain, continuous output space

Empirical Risk Minimization

Exercise 4.3: From Risk to Metrics

optional

Consider binary pedestrian detection in autonomous driving.

1. Write the empirical risk minimization (ERM) objective using cross-entropy loss. Let $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ be the training dataset and ℓ_{CE} the cross-entropy loss. What is $\hat{R}(\theta)$? avg over dataset
2. Why do we optimize cross-entropy loss instead of directly optimizing the metric we care about (e.g., recall on pedestrians)? because we can't calculate a derivative on the metrics and max func is not differentiable
need a surrogate loss function
3. If a model has low training loss (high accuracy) but low recall on the pedestrian class, what might explain this? (Hint: think about class imbalance and what accuracy actually measures.) if the positive cases of pedestrians on street are underbalanced/unrepresented the predictor will learn to only predict false with results in good enough acc/loss but bad recall and it will never trigger in an hazard
4. How would you detect this problem in practice? What metrics would you compute?
edge case and hazard tests, recall, class balance by data distribution.

Distribution Shift and Data Quality

Exercise 4.4: Distribution Shift Types

Consider the following three scenarios involving the CARLA autonomous driving model (trained on sunny daytime data):

1. The model is deployed during winter, when roads are often wet and the sun is at a low angle, creating glare in camera images. covariant shift real use case different, goal the same
bad metrics over all -> unusable, new training data
2. A new city zone is added to the test route where 60% of road users are cyclists - a class that was rare (< 5%) in the original training data. label shift: different test representation: with enough training this should still work perfectly, class balancing
3. The city replaces all old traffic light housings with a new, slimmer design that the model has never seen. concept shift goal changes to test: uncertainty goes up, but model should still work, different types of traffic signs in training for generalization

For each scenario:

- Identify which type of distribution shift applies (covariate shift, label shift, or concept shift). Justify your answer.
- Explain what effect you would expect on model performance.
- Suggest one mitigation strategy.

Exercise 4.5: ODD Coverage with k -Projection Coverage

You defined an ODD in Exercise Sheet 2. Now assess how well your *test set* covers this ODD using k -projection coverage. skipped because of compute limit and recalculation

- Describe in your own words what k -projection coverage measures and why it is a useful metric for ODD coverage. it measures whether combinations of data attributes are present in the dataset and how many different usecases need to be tested
- Download the implementation from <https://github.com/kkirchheim/odd-coverage> and compute the k -projection coverage of your test set for $k \in \{1, 2, 3\}$. impl, but not applied
- How does coverage change with increasing k ? What does this tell you about the adequacy of your test set?

depends on the OOD: coverage for small OOD specification. diverse sets will be more difficult

Practical: Testing the CARLA Model

Exercise 4.6: Designing a Test Suite from Safety Constraints

Refer back to Exercise 2.7 in Sheet 2, where you derived safety constraints from Unsafe Control Actions (UCAs). For each **model-level constraint**, design **one test case** using the following structure:

Constraint ID	Constraint	Test input description	Expected output	Pass criterion
9	pedestrian detection must not miss any pedestrians standing in front	pic, pedestrian in front	acc, recall	recall above 0.997
5	OOD restrictions call human for action	take over call time after IOD switch	time in ms	OOD detection recall above 0.98
6	emergency breaking not initiated with low confidence	pictures of false positives	acc, recall, f1, confidence	precision above 0.9 uncertainty below 0.3
8	emergency breaking also detects animals	pictures of animals on street	acc, recall	ped acc above 0.9 for animals recall above 0.95

Hint: Include the constraint ID (e.g., SC-1, SC-3, SC-4) in the first column. This links your test design explicitly to the safety constraints you identified.

Exercise 4.7: Per-Class Evaluation

Evaluate each of your three models on the provided test set.

- Compute and report precision, recall, and F1-score for each model. see pictures
- Plot a confusion matrix for each model. see pictures
- Which model has the lowest recall? Is this the model you expected to be hardest based on your hazard analysis?

yes, the pedestrians are to underrepresented

4. Based on a safety argument: what minimum recall would you require for the *pedestrian* model before considering deployment? Justify your threshold.

0.997 -> better than human at least